

Yu Li

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Education

George Washington University	Washington, D.C.
<i>Ph.D. Candidate in Electrical and Computer Engineering, GPA: 4.0/4.0</i>	<i>2025.08 – 2029.05</i>
Advisor: Prof. Tian Lan 📧; Work with: Prof. Zhengling Qi 📧	<i>(expected)</i>
Wuhan University, Hongyi Honor College	Wuhan, China
<i>B.Eng. in Microelectronics Science and Engineering, GPA: 3.87/4.0</i>	<i>2021.09 – 2025.06</i>

Experience

ByteDance, Tiktok	San Jose, CA
<i>Research Scientist Intern, Recommendation Core</i>	<i>2026.05 – 2026.08</i>
- Owned the ranking stage of TikTok video recommendation, built on a 3B-parameter Large Recommendation Model. - Adapted LLM architecture techniques to the ranking model, including multi-head latent attention, time-aware RoPE, and mixed-query optimization. - Carried the model from offline training to production deployment, with gains on core business metrics.	

Research Highlights

My research develops principled reinforcement learning formulations for **LLM Reasoning**, spanning token-level credit assignment, preference optimization, and context conditioning [NeurIPS'26] [NeurIPS'26][P2][P3], and extends these formulations to **Agentic RL** through hierarchical skill acquisition and multi-turn policy optimization [P1][ACL'26][INFOCOM'26]. On **Efficient AI**, my work focuses on parameter-efficient fine-tuning and long-context KV cache management [COLM'26][AAAI'26]. Earlier work on **Generative Models** covers diffusion-based personalized content-style synthesis [CVPR'26].

Preprints & Under Review

P=Preprint

- [P1] **ARISE: Agent Reasoning with Intrinsic Skill Evolution in Hierarchical Reinforcement Learning** 📧
Yu Li, Rui Miao, Zhengling Qi, Tian Lan
- [P2] **Right Meets Wrong: Bilateral Context Conditioning with Reward-Confidence Correction for GRPO** 📧
Yu Li, Tian Lan, Zhengling Qi
- [P3] **Hindsight Self-Distillation: Demonstrating to the Teacher for On-Policy LLM Reasoning** 📧
Yu Li, Shu Hong, Tian Lan
Under review. ARR May 2026 OA: 3.33/5

Selected Publications [Full List]

†=Equal Contribution

- [NeurIPS'26] **OPPO: Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning**
Yu Li, Rui Miao, Tian Lan, Zhengling Qi

- [[NeurIPS'26](#)] **INSPO: Unlocking Intrinsic Self-Reflection for LLM Preference Optimization**
Yu Li, Tian Lan, Zhengling Qi
- [[COLM'26](#)] **MomentKV: Closing the Directional Gap in KV Cache Eviction for Long-Context Inference**
Yu Li, Binxu Li, Tian Lan
- [[ACL'26 Findings](#)] **Reason in Chains, Learn in Trees: Self-Rectification and Grafting for Multi-turn Agent Policy Optimization**
Yu Li, Sizhe Tang, Tian Lan
- [[CVPR'26](#)] **CRAFT-LoRA: Content-Style Personalization via Rank-Constrained Adaptation and Training-Free Fusion**
Yu Li, Yujun Cai, Chi Zhang
- [[ICASSP'26 Oral](#)] **KG-SAM: Encoding Structural Constraints into Segment Anything Models via Probabilistic Graphical Models**
Yu Li, Da Chang, Xi Xiao
- [[INFOCOM'26](#)] **ACDZero: Graph-Embedding-Based Tree Search for Mastering Automated Cyber Defense**
Yu Li[†], Sizhe Tang[†], Rongqian Chen, Fei Xu Yu, Guangyu Jiang, Mahdi Imani, Nathaniel D. Bastian, Tian Lan
- [[ACM MM'26](#)] **Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers**
 Yuqi Li, Xi Xiao, Yunbei Zhang, Lin Zhao, Yu Li, Aiden Zhao, Tianyang Wang, Hao Xu, Yingli Tian
- [[AAAI'26](#)] **Calibrating and Rotating: A Unified Framework for Weight Conditioning in PEFT**
 Da Chang, Peng Xue, Yu Li, Yongxiang Liu, Pengxiang Xu, Shixun Zhang

Selected Awards & Honors

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- **ICML 2026 Gold Reviewer**, top reviewers recognition. 2026
 - **Innova International Exchange Scholarship**, 6 recipients university-wide, ¥70,000. 2024
 - **Innova Excellence Scholarship**, Top 3%, ¥10,000, twice. 2023, 2024
 - **First-Class Scholarship**, Top 5%, ¥3,000, three times. 2022, 2023, 2024

Academic Services

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- **Conference Reviewer:** 2027: AAI. 2026: NeurIPS, ICML, ICLR, COLM, AAI, ACM MM.
 - **Journal Reviewer:** TPAMI, TMLR, ToN, Neurocomputing.

Skills

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- **Training Frameworks:** PyTorch, verl, Hugging Face Transformers / Accelerate / TRL, Hydra.
 - **Inference & Distributed:** vLLM, Ray, FSDP, NCCL multi-node training, Docker, Linux GPU clusters (SSH, tmux).
 - **Languages:** English (Proficient), Mandarin (Native), Japanese (Basic).